

Addendum to pre-registration form

Chromatic discrimination in the lower and upper visual fields

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This addendum supplements the pre-registration document dated 22 June 2026. It does not revise or replace any part of that document; its sole purpose is to record, before we draw any confirmatory conclusions, a decision about sample size that we reached after examining the data from the first three participants.

What we observed in the initial three participants

We have now collected and analysed data from the three experienced participants described in the original document. For two of the three participants the measurements were internally consistent, the Wishart Process Psychophysical Model (WPPM) fits were well constrained in both the upper and the lower visual field conditions, and the estimated discrimination ellipsoids differed between the two field locations in a manner consistent with Hypothesis 1. Ellipsoids from the upper and lower visual fields were similar in shape but sensitivity was higher in the lower visual field for these participants, and their data were similar to each other.

For the third participant, the shape of upper and lower visual field ellipsoids was different. We suspect that adaptive trial placement may not have proceeded as intended: trials were not well distributed with respect to the fit threshold surface, more so for the lower visual field, and the resulting WPPM fits appeared sub-optimal. Poor trial placement could produce poor threshold ellipsoid estimates. A feature of the outlier participant's trial placement in the lower visual field is that it was asymmetric around the reference. If the participant's underlying isodiscrimination surface is in fact similarly asymmetric, it would not be well fit by the WPPM in its current form and this is also a possibility. This a substantial individual difference from the other two participants. We therefore regard the data from this participant as inconclusive and do not draw any firm inferences from them in their present form.

Decision

The original document stated that we viewed it as “reasonably likely that we will decide to collect additional data beyond that described here from the initial three experienced participants”. Having examined the initial data, we have now made that decision. Specifically, we will:

1. recruit and test 4 additional participants, bringing the planned total to 7 participants; and

2. re-collect the data from the third participant described above.

The purpose of collecting these additional data is to establish more clearly whether the upper–lower difference suggested by the first two participants is reliable across observers, as well as to better understand the behavior and effect of trial placement for this experiment.

Additional participants

The additional participants will satisfy the eligibility and screening criteria set out in the original document: aged 18 years or over, normal or corrected-to-normal visual acuity, normal colour vision as assessed by standard colour vision testing (e.g. Hardy-Rand-Rittler plates), and written informed consent under the ethics protocol approved by the Central University Research Ethics Committee (CUREC) at the University of Oxford. Compensation remains at the approved rate of £12 per hour. In contrast to the initial three participants, the additional participants are not required to be experienced psychophysical observers.

No other changes

No other aspect of the pre-registration is changed. The hypotheses, apparatus, stimuli, procedure, nominal number of trials per visual-field location, and the planned analyses — WPPM estimation of the three-dimensional discrimination ellipsoid, bootstrapped 95% confidence intervals, and the comparison between the upper and lower visual fields — all remain as described in the original document. The additional and re-collected data will be obtained under that same protocol, and the data already collected from the first two participants will be included in the final analysis.